

A practical beginner guide to the STACK API

Sam Fearn

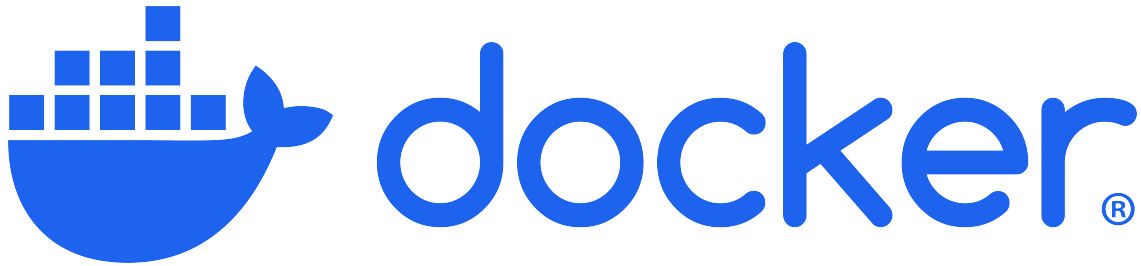
Durham University

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Aims

- Explain the basics of what the STACK API is/does.
- Workshop converting a sample tex file into html with an embedded example stack question (follow-along).
- We will use Docker Desktop (`https://www.docker.com/products/docker-desktop/`).



Moodle Quizzes

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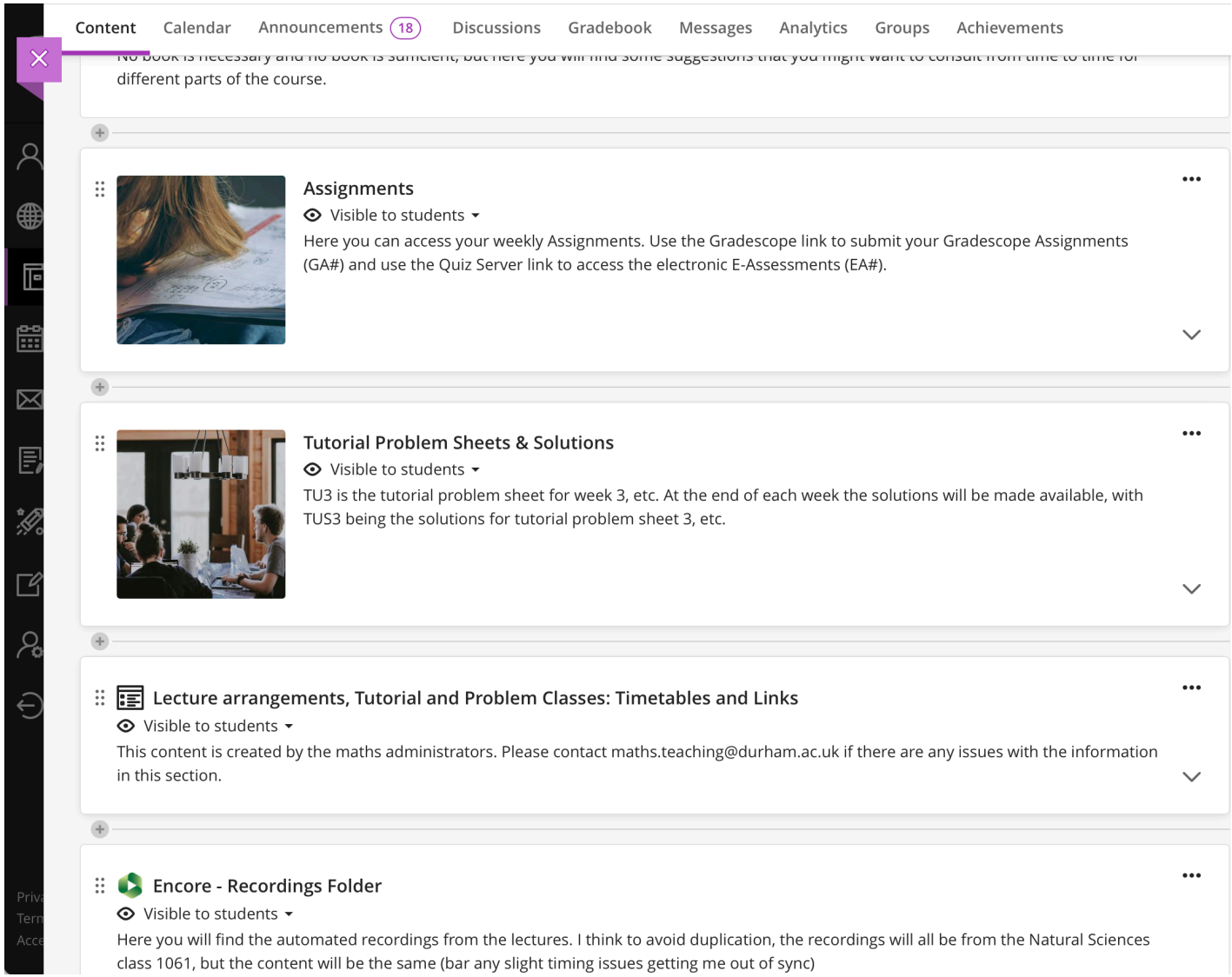
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Most of these features are likely necessary for *summative* assessment (assessment *of* learning), but may not be for *formative* assessment (assessment *for* learning).

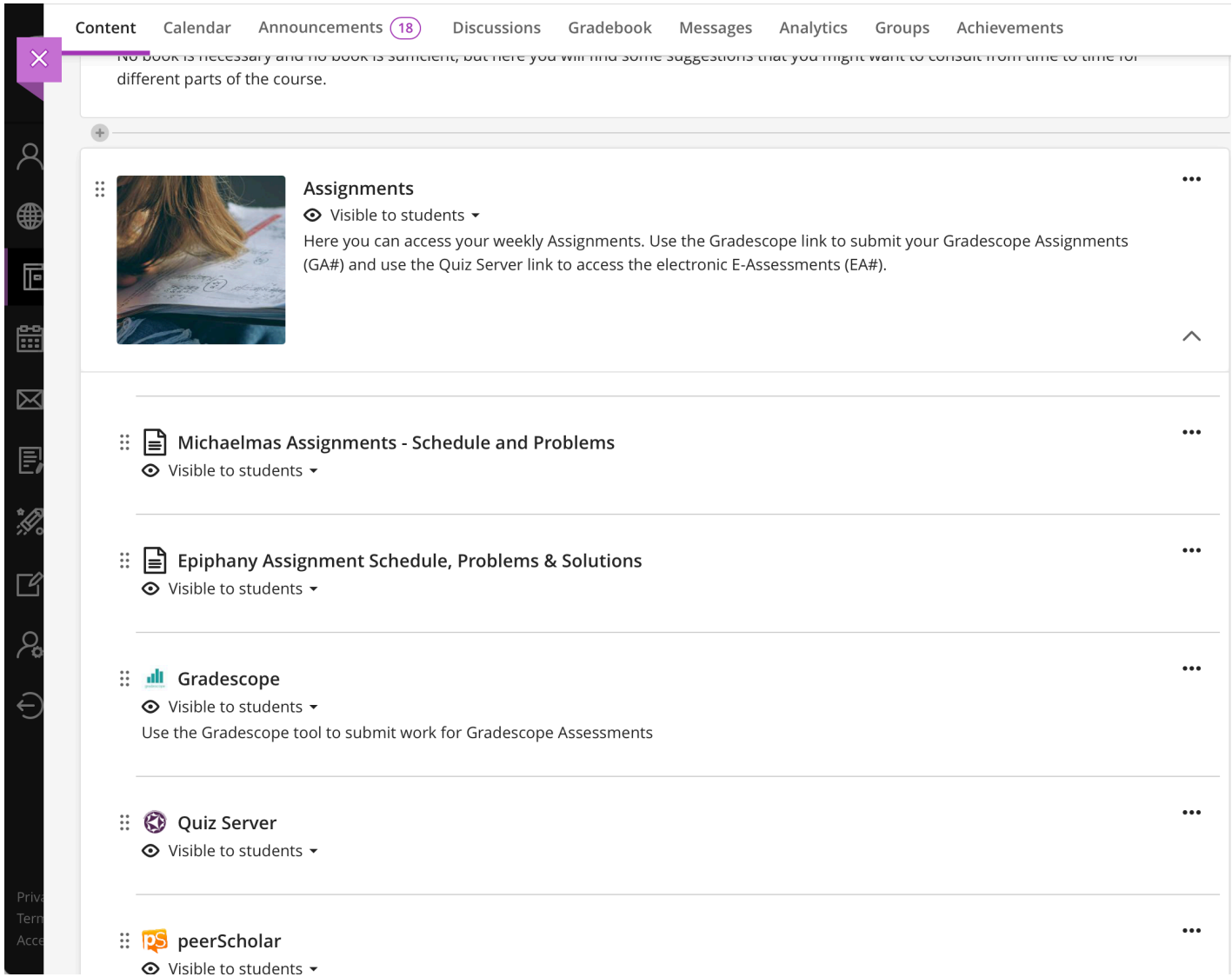
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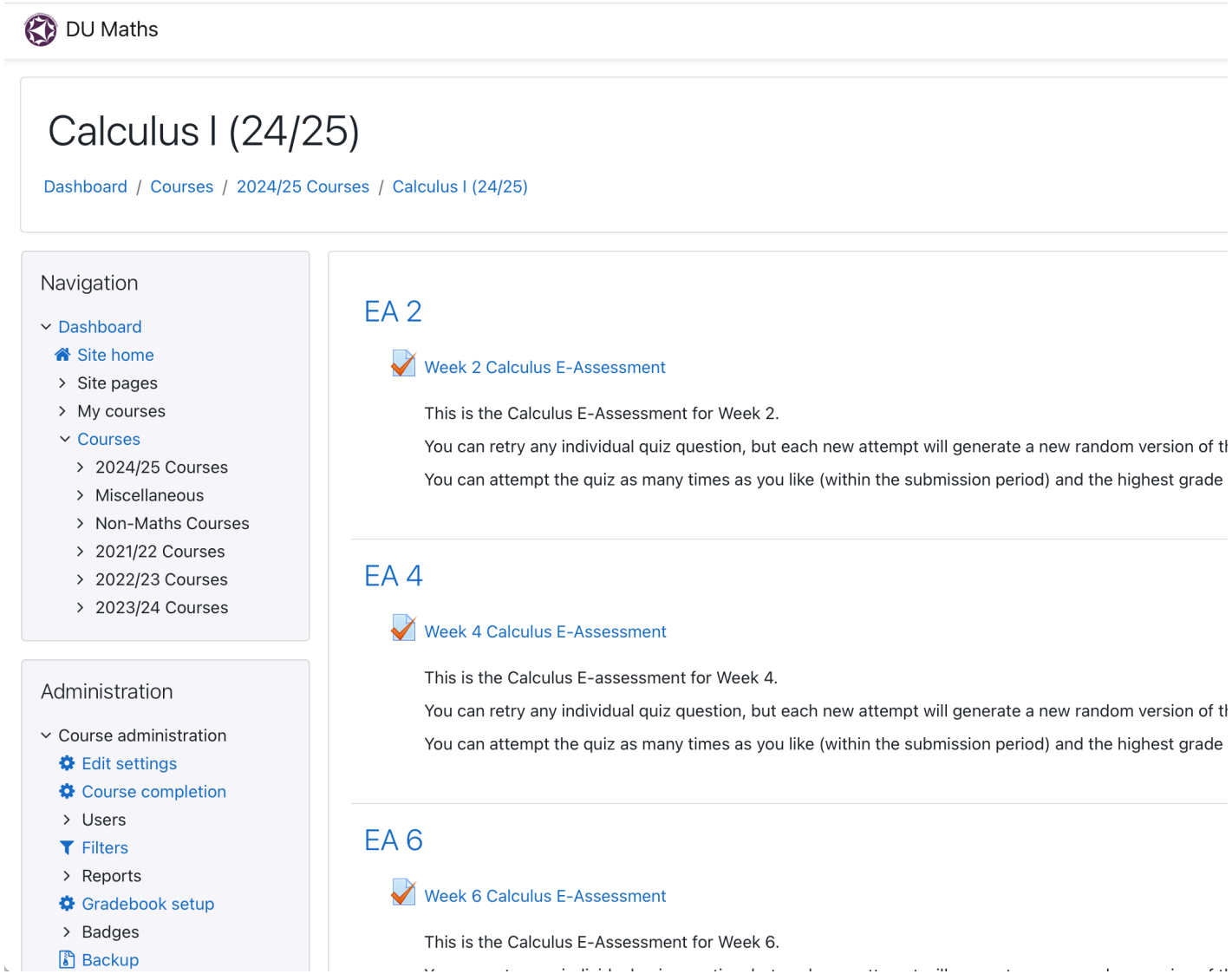
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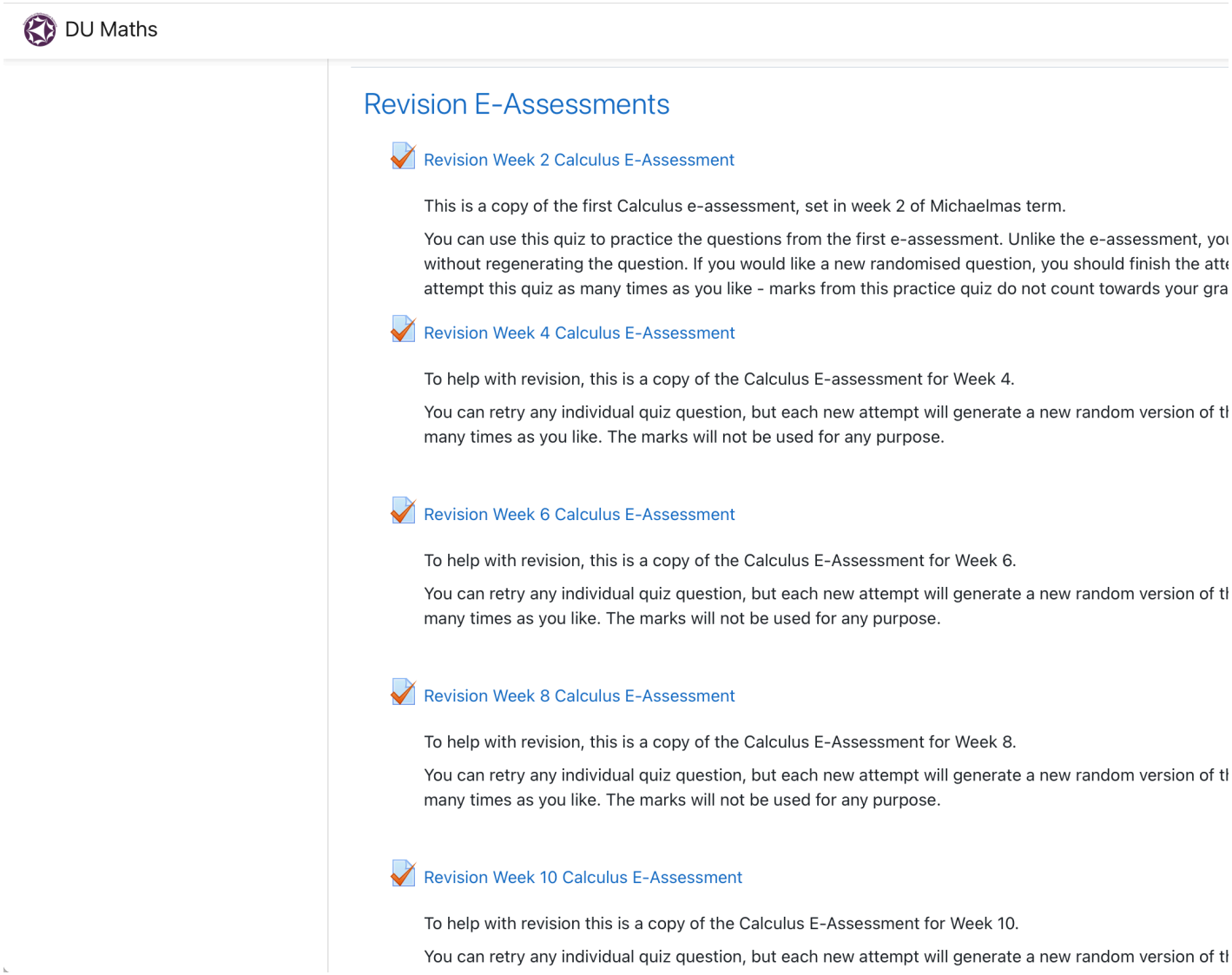
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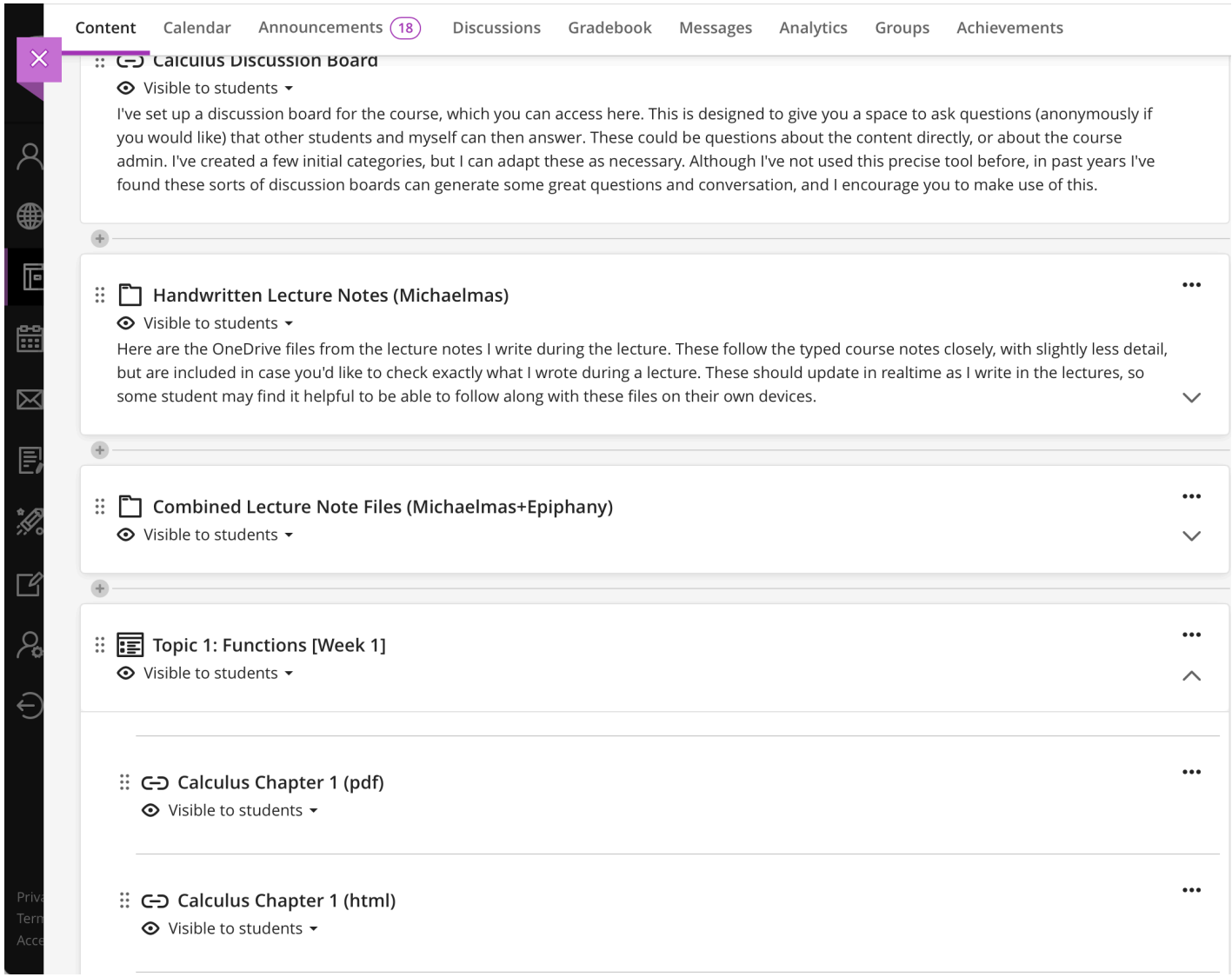
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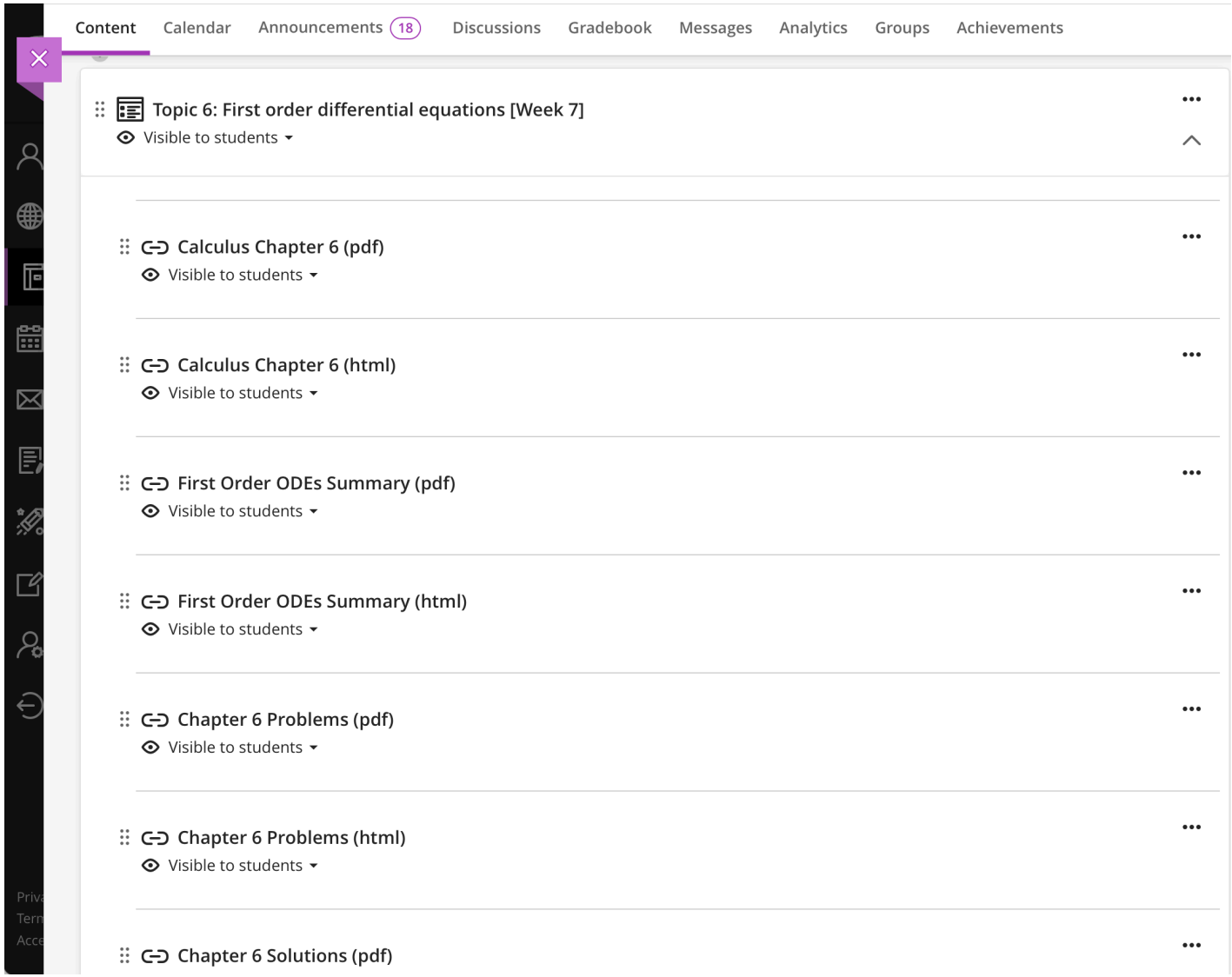
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6.5 Bernoulli equations

A **Bernoulli equation** is a nonlinear ODE of the form

$$y' + p(x)y = q(x)y^n$$

where $n \neq 0, 1$, otherwise the equation is simply linear.

These ODEs can be solved by first performing the substitution $v = y^{1-n}$, which converts the equation to a linear ODE for $v(x)$.

Eg. Solve

$$y' - \frac{2y}{x} = -x^2y^2.$$

This is a Bernoulli equation with $n = 2$ hence put $v = \frac{1}{y}$, which gives $v' = -\frac{y'}{y^2}$.
Dividing the ODE by y^2 yields

$$\frac{y'}{y^2} - \frac{2}{xy} = -x^2$$

which in terms of v is

$$-v' - \frac{2v}{x} = -x^2$$

ie the linear equation

$$v' + \frac{2}{x}v = x^2.$$

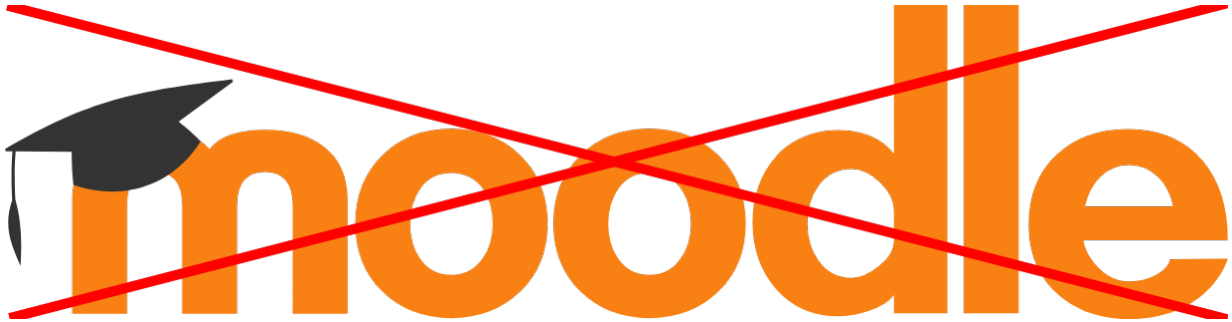
We solve this with an integrating factor $I = \exp\left(\int \frac{2}{x}\,dx\right) = \exp(2\log x) = x^2$ as

$$v = \frac{1}{x^2}\int x^2x^2\,dx = \frac{1}{x^2}\left(\frac{1}{5}x^5 + \frac{c}{5}\right).$$

So finally

$$y = \frac{1}{v} = \frac{5x^2}{x^5 + c}.$$

Questions in notes



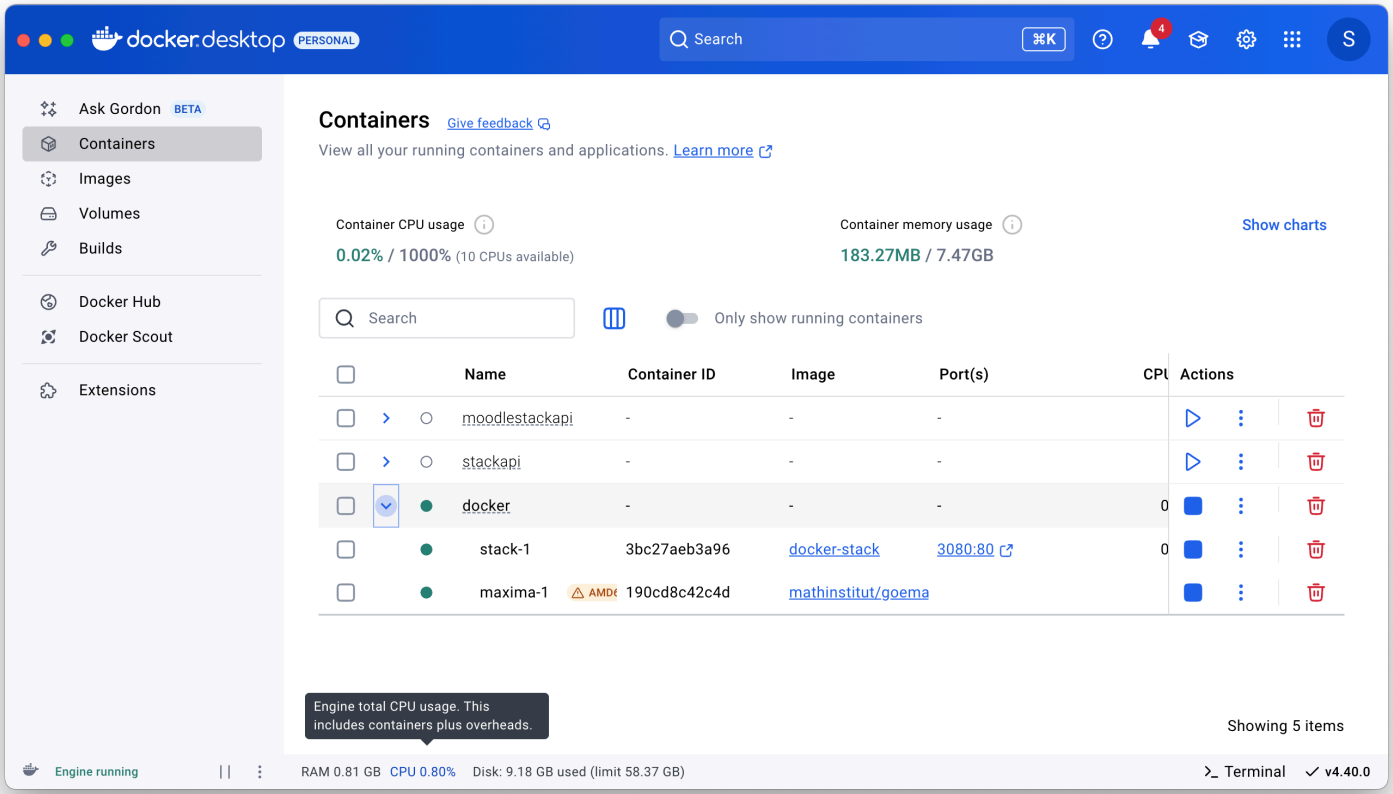
The STACK API allows questions to be used outside of Moodle, allowing for the possibility of embedding STACK questions directly inside html lecture notes.

DEMO

https://github.com/samfearn/stack-api-seminar

Setup

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Each route expects to receive a POST request, whose body is a JSON object containing the question XML (and possibly other fields).

Each route responds with a JSON object.

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  If going straight to html, the following can instead be inserted into the html directly:

  ```
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- The html notes and question XML file/s (as well as an additional 'stackapicalls.js') are hosted on an appropriate webserver.

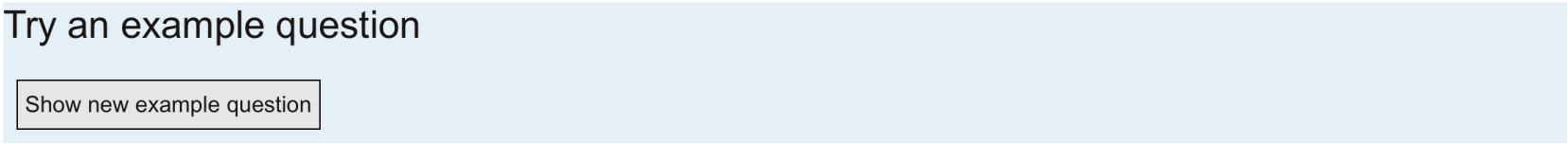
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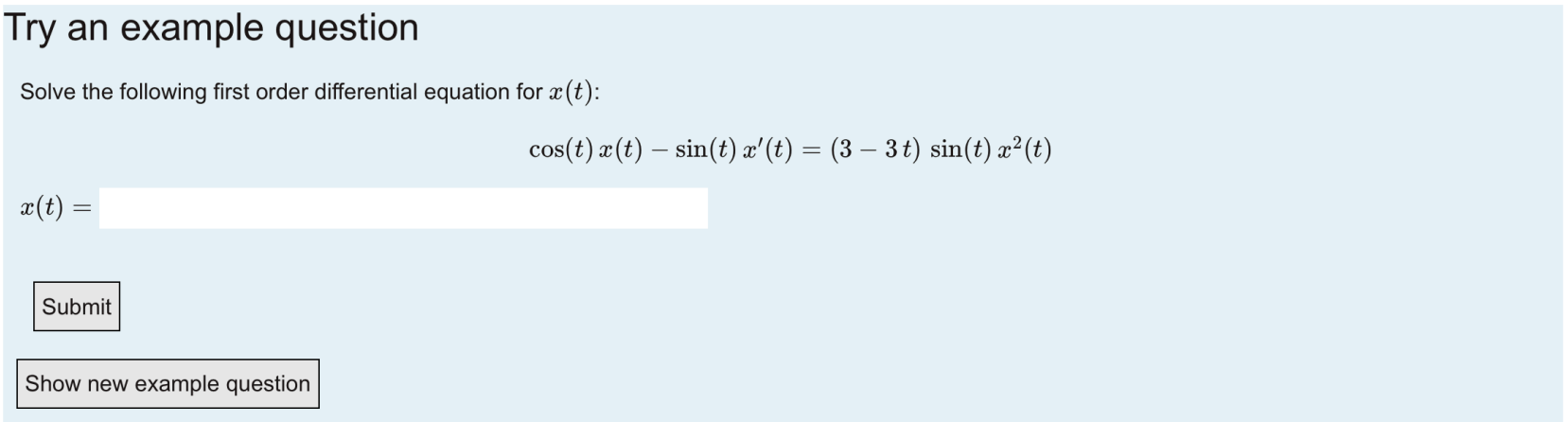
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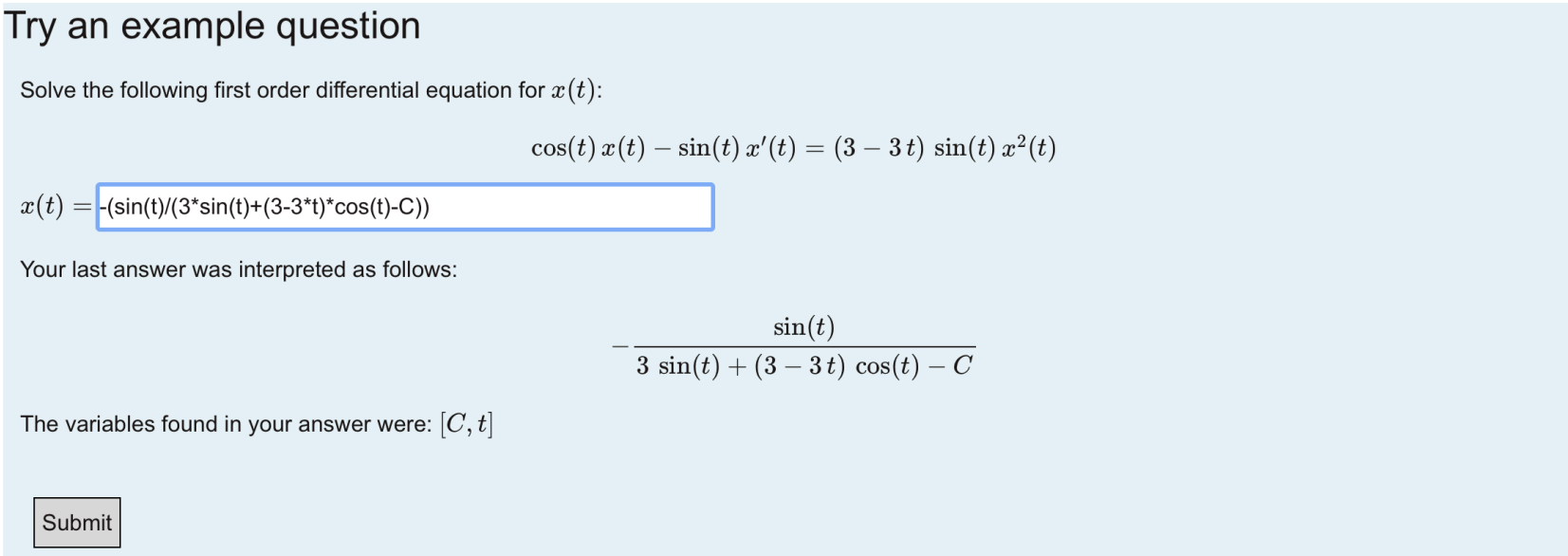
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- Event listeners are attached to the inputs to call the /validate route when the input changes.
- A random seed is used for the render, so the seed is attached to the answer submit button.
- The answer submit button calls the /grade route, passing the xml as before, with the matching seed to the /render call. The response from this is then rendered to the page.

Process - key steps

$$-\frac{\sin(t)}{3\,\sin(t)+(3-3\,t)\,\cos(t)-C}$$

The variables found in your answer were: $[C, t]$

Correct answer, well done.

- Your answer satisfies the differential equation.
- Your general solution contains an unknown constant as expected.

Marks for this submission: 1.00 / 1.00.

Submit

General feedback:

This is a Bernoulli differential equation. To solve this question, you should first write the ODE in the form

$$x^{-n}\frac{dx}{dt}+p(t)x^{1-n}=q(t),$$

for some n. If you then make the substitution $v=x^{1-n}$, then this ODE becomes a linear ODE in terms of v. Once you've solved this ODE for v, you then need to use the substitution in order to find the required solution for x.

Score: 1.00 out of 1

Show new example question

Notes and caveats

- Currently, the routes expect the question xml to be sent as part of the request. This means a knowledgeable user can obtain the full question xml.

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- Similarly, the response from /render contains the model answer. This is useful for a 'reveal answer/s', even if the general/specific feedback doesn't contain this.
- In the example, the questions are essentially always in 'Adaptive mode'.

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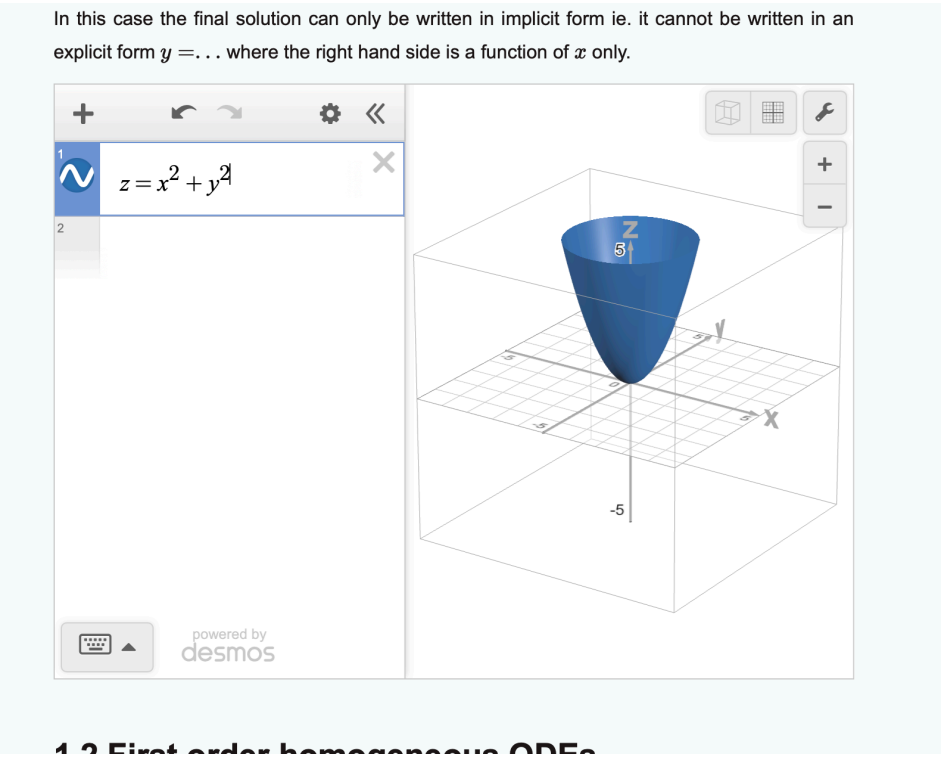
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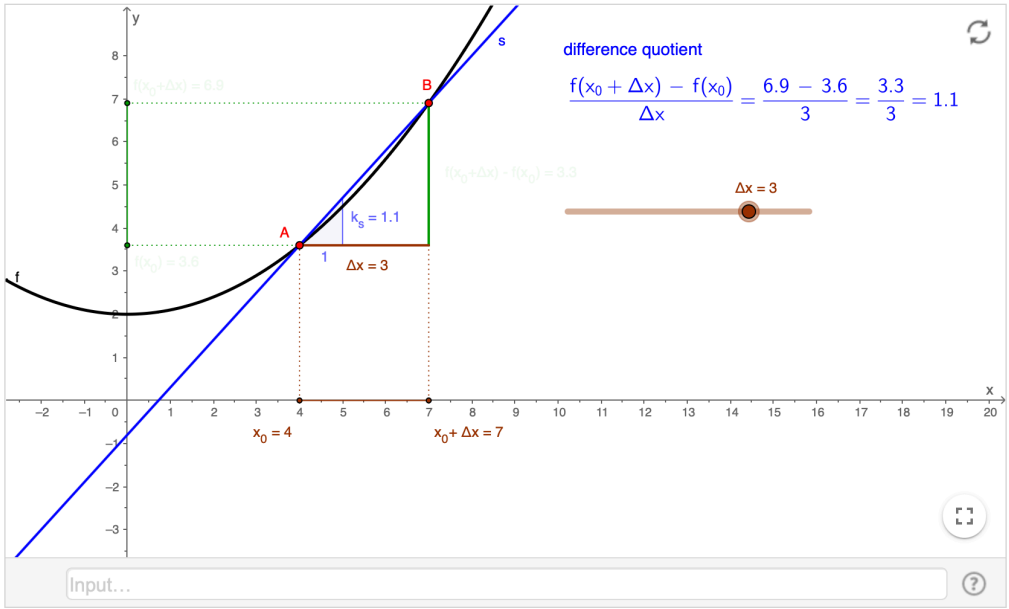
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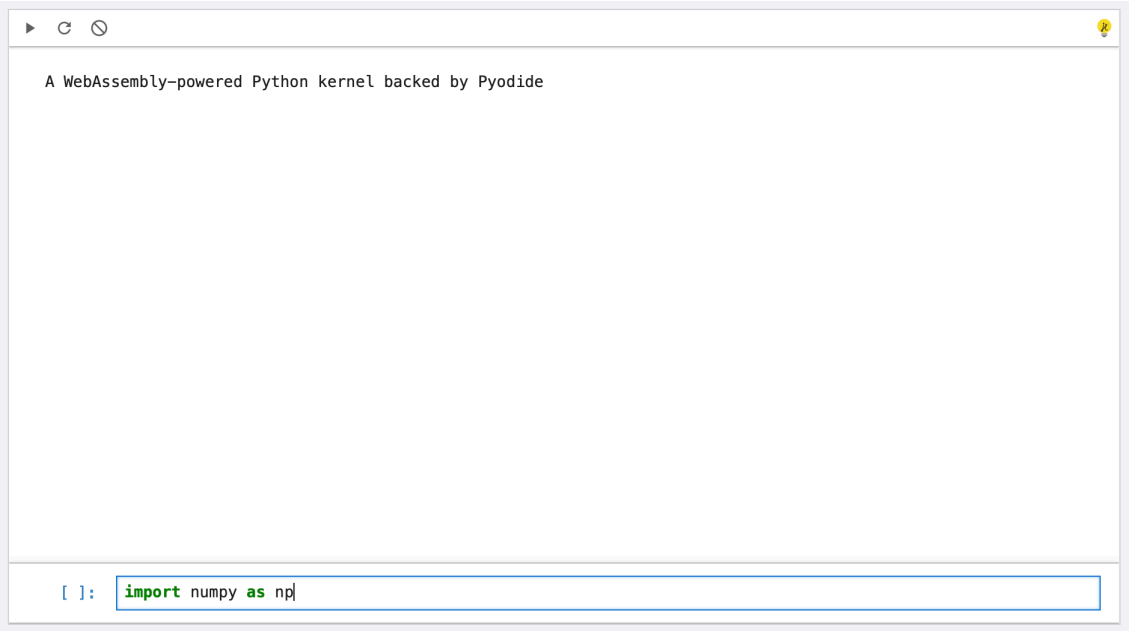
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Andreas Lindner

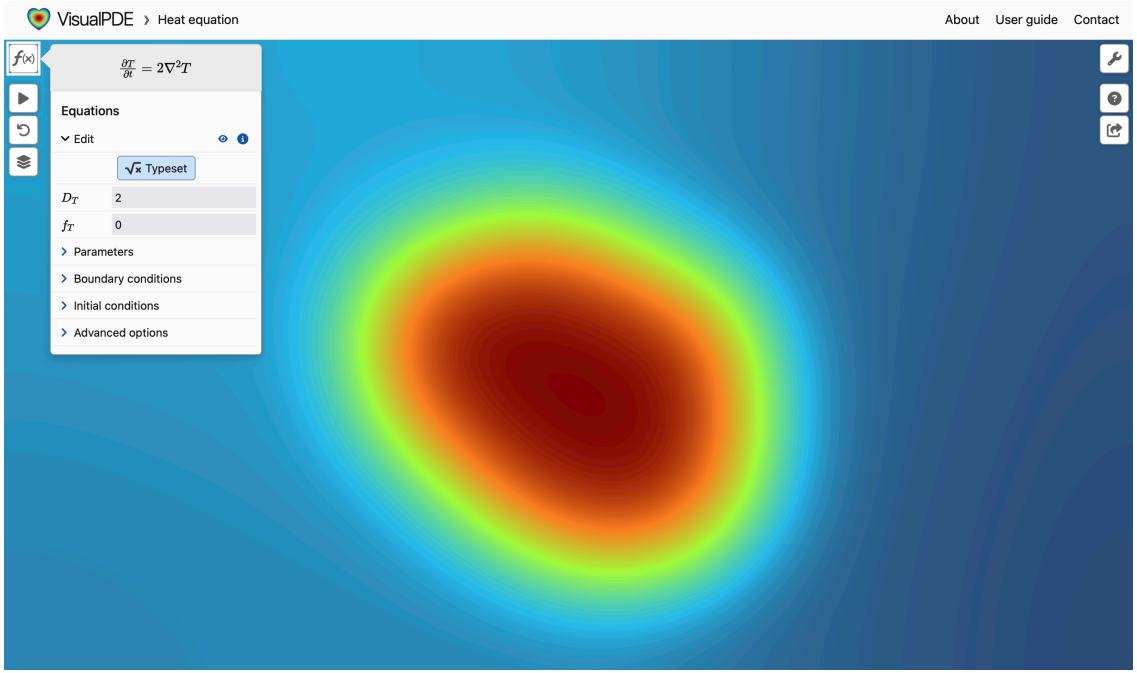
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- We can also embed other interactive elements: Desmos, Geogebra, JupyterLite, Visual PDE, ...
- While formal assessment questions are good to include, there is also (massive) scope for additional questions (and proofs), and more scaffolded examples.

Thanks for listening!

Any Questions/Comments?